

Generative AI & LLMs

Assignment-6 Report

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Introduction:

- Generative modeling aims to learn the underlying data distribution such that new, realistic samples can be generated. This project compares three generative models of increasing complexity on the MNIST dataset: a Standard GAN, a VAE-GAN, and a VAE-WGAN.
 - 1. The **Standard GAN** formulates generation as a two-player minimax game between a Generator and a Discriminator. While it can generate sharp images, it often suffers from training instability, vanishing gradients, and mode collapse.
 - 2. The **VAE-GAN** combines the structured latent space of a Variational Autoencoder with the perceptual realism of a GAN. The encoder enforces a continuous latent distribution using KL divergence, while adversarial training improves sample sharpness.
 - 3. The **VAE-WGAN** further improves training stability by replacing the standard GAN loss with the Wasserstein distance. By enforcing a Lipschitz constraint on the critic, the model provides smoother gradients and more stable convergence.
 - Each successive model aims to address the limitations of the previous one, improving stability, diversity, and latent space structure.
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Experimental Setup:

1. Dataset:

- MNIST (28×28 grayscale)
- Normalization: In the range $[-1,1]$
- Batch size: 128

2. Architecture Description

1. GAN architecture:

a. Generator :

Input (256*7*7) → RELU → Unflatten layer (256,7,7) → ConvTranspose2D → RELU → ConvTranspose2D → Tanh (output)

b. Discriminator : Input $\rightarrow$ Conv2d $\rightarrow$ LeakyReLU $\rightarrow$ Conv2D $\rightarrow$ LeakyReLU $\rightarrow$ Flatten $\rightarrow$ Dense $\rightarrow$ Sigmoid

2. VAE-GAN architecture:

a. VAE encoder : Input $\rightarrow$ Conv2D $\rightarrow$ Relu $\rightarrow$ Conv2D $\rightarrow$ ReLU $\rightarrow$ Flatten $\rightarrow$ mean dense layer $\rightarrow$ log(variance) dense layer.

b. Generator / VAE decoder : Input (256*7*7) $\rightarrow$ RELU $\rightarrow$ Unflatten layer (256,7,7) $\rightarrow$ ConvTranspose2D $\rightarrow$ RELU $\rightarrow$ ConvTranspose2D $\rightarrow$ Tanh (output)

c. Discriminator : Input $\rightarrow$ Conv2d $\rightarrow$ LeakyReLU $\rightarrow$ Conv2D $\rightarrow$ LeakyReLU $\rightarrow$ Flatten $\rightarrow$ Dense $\rightarrow$ Sigmoid

d. LOSS = Binary CrossEntropy

3. VAE-WGAN architecture:

a. VAE encoder : Input $\rightarrow$ Conv2D $\rightarrow$ Relu $\rightarrow$ Conv2D $\rightarrow$ ReLU $\rightarrow$ Flatten $\rightarrow$ mean dense layer $\rightarrow$ log(variance) dense layer.

b. Generator / VAE decoder : Input (256*7*7) $\rightarrow$ RELU $\rightarrow$ Unflatten layer (256,7,7) $\rightarrow$ ConvTranspose2D $\rightarrow$ RELU $\rightarrow$ ConvTranspose2D $\rightarrow$ Tanh (output)

c. Discriminator : Input $\rightarrow$ Conv2d $\rightarrow$ LeakyReLU $\rightarrow$ Conv2D $\rightarrow$ LeakyReLU $\rightarrow$ Flatten $\rightarrow$ Dense $\rightarrow$ Sigmoid. d. Loss= Wasserstein Distance

3. Table: Key Differences

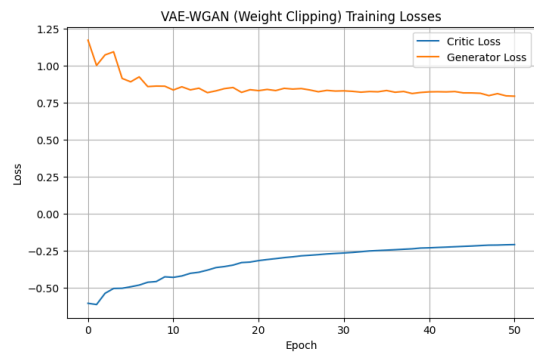
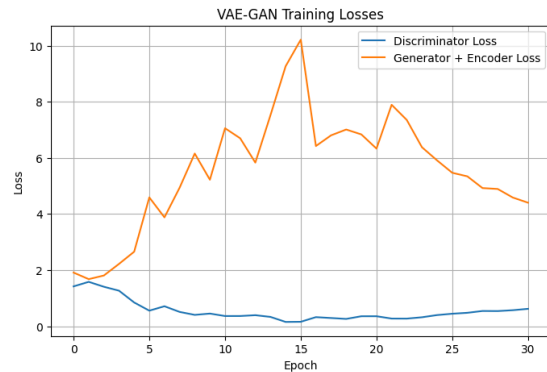
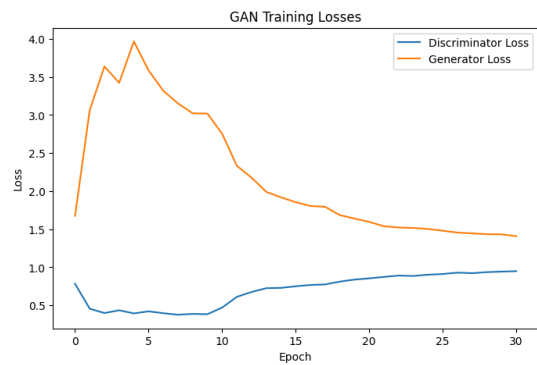
Model	Loss Function	Optimizer	Constraint	Latent Space
GAN	BCE (JS Divergence)	Adam (0.0002)	None	Unstructured
VAE-GAN	BCE + KL + Feature Loss	Adam (0.0002)	KL Regularization	Continuous
VAE-WGAN	Wasserstein + KL + Feature Loss	RMSProp / Adam (0.00005)	Weight Clipping	Continuous

4. Training Details

- Epochs = 50
- Latent space Dimension = 100
- n_critic: 5
- Clip value: 0.01
- KL weighting: $\beta = 0.1$

Results & Comparison

A. Training Stability:

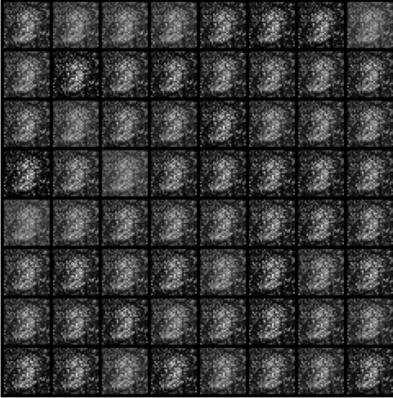


Observations:

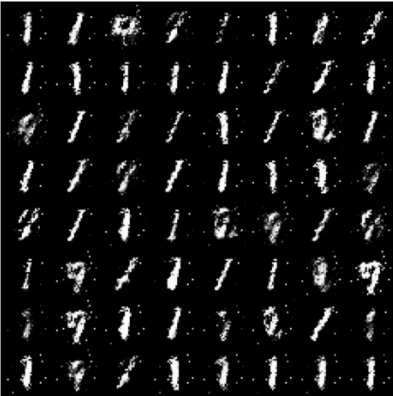
- VAE-WGAN has most stable training because it smoothly converges loss to minimal which indicates stable training

B. Quality of Generated Samples:

GAN Epoch 0



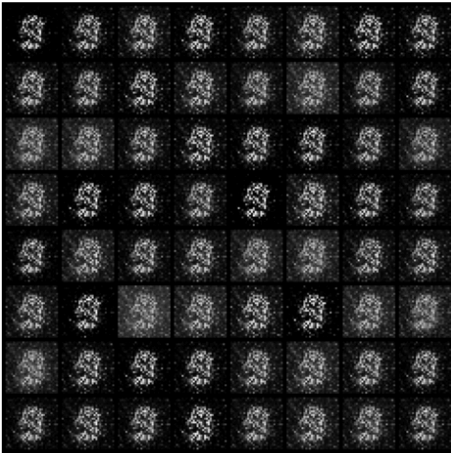
GAN Epoch 10



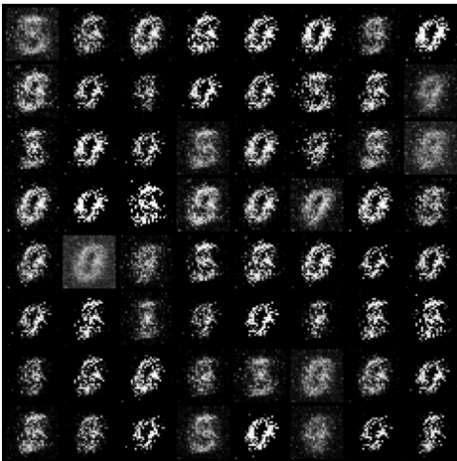
GAN Epoch 30



VAE-GAN Epoch 0



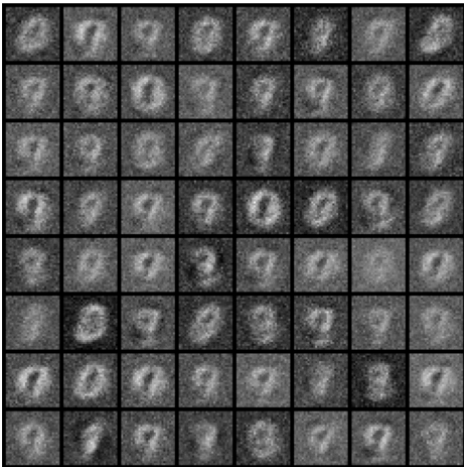
VAE-GAN Epoch 10



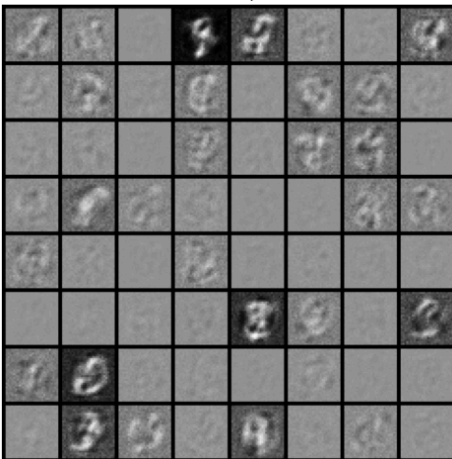
VAE-GAN Epoch 30



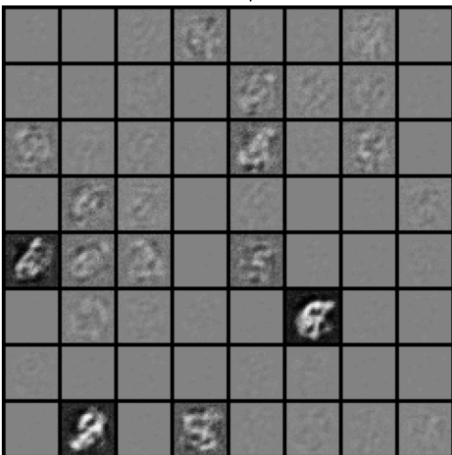
VAE-WGAN Epoch 0



VAE-WGAN Epoch 20



VAE-WGAN Epoch 40



C. Latent Space Interpolation:

Latent Interpolation (VAE-WGAN)



The WGAN produces the most semantically meaningful transitions because of several key properties:

1. Continuous and structured latent space – the combination of VAE and GAN ensures the latent space is smooth and meaningful.
2. Reduced mode collapse – using the Wasserstein distance as the loss encourages the generator to cover the full data distribution.
3. Stable training – WGAN training is more stable compared to standard GANs.
4. Avoidance of vanishing gradients – enforcing the Lipschitz constraint via weight clipping helps maintain useful gradients for the generator.

Overall, the structured latent space comes from integrating VAE with GAN, mode collapse is mitigated by the Wasserstein loss, and vanishing gradients are prevented through weight clipping.

Conclusion:

- In my observations, all the three models, GAN, VAE GAN were found to generate good and clarity images but VAE WGAN produced noisy images for the taken epochs
- But when compared the loss curves of the above three, VAE WGAN has shown less oscillations, so it might have given less noisy images if tried for more epochs
- With VAE WGAN, we had continuous and structured latent space too.